

Moving State Estimation by CNN from Long Time Data of Smartphone Sensors

Sussex-Huawei Locomotion Challenge 2023

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ABSTRACT

This paper presents the ideas utilized by *Team Ds* at the 2023 Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge. The 2023 SHL Challenge involves classifying eight modes of locomotion and transportation using data acquired by smartphone motion and GPS sensors. The data were obtained from three users, User1 provided the training set, and parts of the data acquired by User2 and User3 were used as the validation and test sets. Long-period data (up to 110 s) were used to estimate the modes of locomotion and transportation for a sample in the main processing. We have created a model to estimate the modes of locomotion and transportation using User1's accelerometer, gyro sensor, magnetic sensor, and GPS location data. The neural network model comprised four sections, i.e., two types of a one-dimensional convolutional neural networks (1D-CNNs) with different filter size using raw waveform data for nine sensors, one 2D-CNN using the spectrograms of accelerometer and gyro sensor, and two fully-connected layers of 22 statistical data computed from the sensors. In a post-processing step, a model was constructed to output the labels from the Soft-Max output obtained from the previous model, and this output was used as the final estimation. The developed model was evaluated experimentally, and the F1-score obtained on the validation data was 0.952.

CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision tasks; Activity recognition and understanding.

KEYWORDS

CNN, Neural network, Long-term data, Multi-input model

ACM Reference Format:

Masaki Kawakatsu, Takumi Hyugaji, and Yugo Toyoshima. 2023. Moving State Estimation by CNN from Long Time Data of Smartphone Sensors: Sussex-Huawei Locomotion Challenge 2023. In *Adjunct Proceedings of the*

2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610749>

1 INTRODUCTION

The SHL Challenge 2023 involves estimating eight modes of locomotion and transportation (1. Still, 2. Walk, 3. Run, 4. Bike, 5. Car, 6. Bus, 7. Train, 8. Subway) based on data acquired from smartphone motion sensors (e.g., accelerometers, gyro sensors, and magnetic sensors) and GPS sensors [1, 2].

The modes of locomotion and transportation is important information that can be used to monitor activity and health, and it provides guidance in terms of selecting appropriate services. In the 2021 SHL Challenge [3], multiple sensors (i.e., GPS location, Wi-Fi reception, GSM reception, and GPS reception) were provided, and different methods were employed to estimate the modes of locomotion and transportation. Team DD [4], which was the top team at the 2021 SHL Challenge, utilized AutoML [5] and additional data from a bus routes dataset, a railway dataset, and a recreational location dataset to improve the modes of locomotion and transportation estimation accuracy.

The second and third place teams, i.e., Team Guerrilla [6] and Team “We_can_fly” [7], used XGBoost [8]. Additionally, the top three teams all utilized multiple time windows. Note that the fourth [9] and fifth [10] ranked teams both used ensemble learning, and their time windows were set to 120 and 5 s, respectively.

We relabeled the labels (8 moving states) every 0.01 s to be one label every 10 s, and all features were treated with the concept of 10-second blocks, which corresponds to the new labeling. This process was applied because 99.84% of the training data and 100% of the validation data have consecutive labels for more than 10 s. Here, we used long-term data (up to 110 s) for the input values in the main processing because such data are less susceptible to outliers. Even if an unexpected behavior occurs during a 10-s period, the characteristics of the remaining 100 s would negate that information. In addition, we believe that using long-term features as input is effective because the labels for this challenge last for approximately 1500 s on average. In this paper, we present the details of our estimation process and the corresponding results.

2 ESTIMATION OF PHONE LOCATION

The sensor data used in this challenge were acquired at four different phone positions (i.e., bag, hips, hand and torso), and the training



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UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico
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ACM ISBN 979-8-4007-0200-6/23/10.
<https://doi.org/10.1145/3594739.3610749>

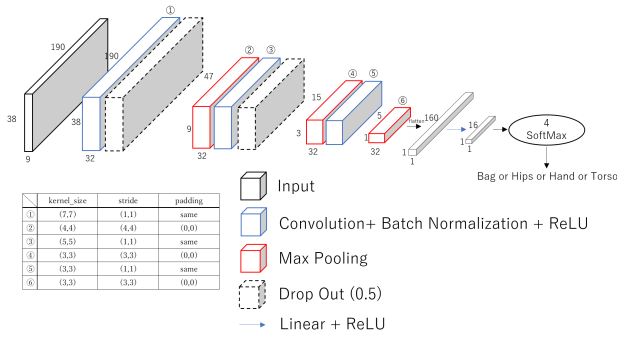


Figure 1: Model structure to estimate holding position.

and validation sets included data for all four positions. However, the test set only included data for a single unknown holding position. Thus, in a preprocessing step, we estimated the phone position of the test set data. The validation and test sets were referred to as User2 and User3 data, respectively, and we generated spectrograms of the accelerometers (x, y, z) at all four phone positions of the validation set for each axis. The (9, 38, 190) image data (sensor, frequency, time), which is the combined spectrogram for 10 s of the block to be predicted and the spectrograms of the two blocks before and after the block, was input to a two-dimensional convolutional neural network (2D-CNN). The model structure is illustrated in Figure 1, and the predicted results are listed in Table 1.

As shown in Table 1, the probability of being the hand position was the highest at approximately 85%. Thus, we estimated the holding position of the test set to be the hand position, and we used only the corresponding hand data to estimate the modes of locomotion and transportation in the subsequent computations.

3 PREPROCESSING

To transform the data into a form suitable for training, the following preprocessing was performed on each feature.

3.1 RELABELING

The labels at every 0.01 s were divided into blocks of 1000 points, and relabeled with the median of the 10 s labels. Here, the last 438 points with fewer than 1000 points have not been used for training. Furthermore, the prediction labels for data with fewer than 1000 points were labeled with the predicted label for the previous block.

3.2 FEATURE ENGINEERING

The input values were created as follows.

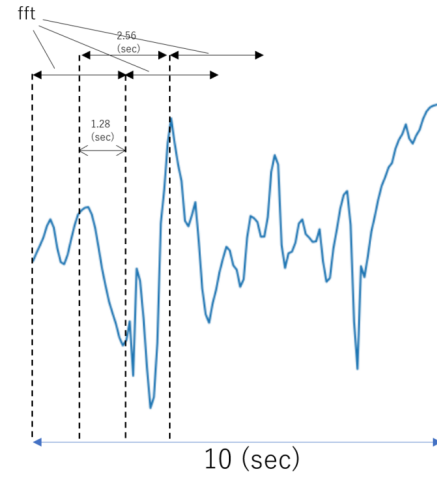


Figure 2: Overview of fast fourier transform (FFT) process.

3.2.1 WAVEFORM DATA FOR 1D-CNN. The raw data of the accelerometers, the gyro sensors, and the magnetic sensors were standardized. To create a nine-channel $\times$ 3000 point (30 s) segment, three contiguous blocks were concatenated.

3.2.2 SPECTROGRAMS FOR 2D-CNN. Six spectrograms from 0–30Hz were created for each block from the standardized raw accelerometers and gyro sensors data. The window size for this process was 256 points, and the overlap was 128 points. Figure 2 shows an overview of the process.

Figure 3 shows examples of the spectrograms created from the accelerometer (x). The spectrograms for five contiguous blocks were concatenated in the time direction to create the image (3 $\times$ 6, 38, 190) used for training.

3.2.3 STATISTICAL DATA FOR FULLY-CONNECTED LAYERS. From the raw data of each sensor, 22 statistical data were computed for each block and then standardized. For the model inputs, we used the 110-s data (11, 22), which is the concatenation of 11 blocks.

The 22 statistical data comprise the following elements.

(1) 10-s average of speed.

The 10-s average of speed is the 10-s average of speed per second calculated from the change of latitude and longitude. The location data used in these calculations were synchronized to 100 Hz data with epoch time. Note that missing values were compensated using linear interpolation. If the calculation resulted in a speed exceeding

Table 1: Percentage of predicted positions.

Position	Prediction (%)
Bag	7.38
Hips	6.30
Hand	85.00
Torso	1.29

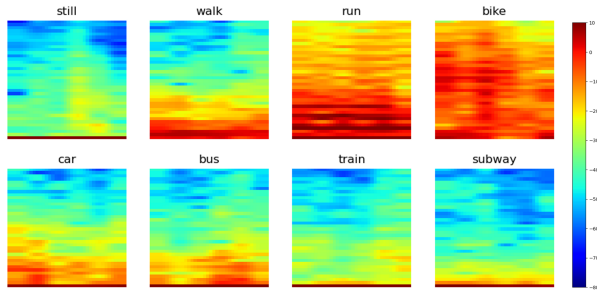


Figure 3: Spectrograms of accelerometer (x).

250 km/h, it was considered an outlier and the value was corrected to precisely 250 km/h.

(2) Number of GPS data acquisitions over 210 s.

(3) 10-s average of altitude.

(4) 10-s average of the standard deviation calculated every 2 s from the magnetic sensor magnitude [11].

(5), (6), (7) 10-s average of accelerometer (x, y, z).

(8), (9), (10) Standard deviation value of accelerometer (x, y, z) for 10 s.

(11), (12), (13) 10-s average value of magnetic sensor (x, y, z).

(14), (15), (16) Standard deviation value of magnetic sensor (x, y, z) for 10 s.

(17), (18), (19) 10-s average value of gyro sensor (x, y, z).

(20), (21), (22) Standard deviation value of gyro sensor (x, y, z) for 10 s.

4 MAIN PROCESSING

A model that predicts the modes of locomotion and transportation from the created features was constructed by concatenating the outputs of the four input sections. An overview of the model is presented in Figure 4. The method of using two types of 1D-CNN models was based on the DeepSleepNet model [12].

Figure 5, Figure 6, Figure 7, Figure 8 show the details of the individual input sections.

5 POSTPROCESSING

A postprocessing model was constructed to make the final prediction. Here, the input to the model was the output of the main processing for 101 blocks (1010 s). The output is the SoftMax output used for the final evaluation. The structure of the postprocessing model is shown in Figure 9.

6 RESULTS

6.1 MAIN PROCESSING RESULTS

The results of the main process were evaluated in terms of accuracy, F1-score, and confusion matrices. The results obtained on the training data, the validation data, and the 100-Hz resampled validation data are shown in Figure 10(a), Figure 10(b), and Figure 10(c), respectively.

As can be seen, the results obtained on the train and validation data demonstrate that “run” state estimation was nearly perfect.

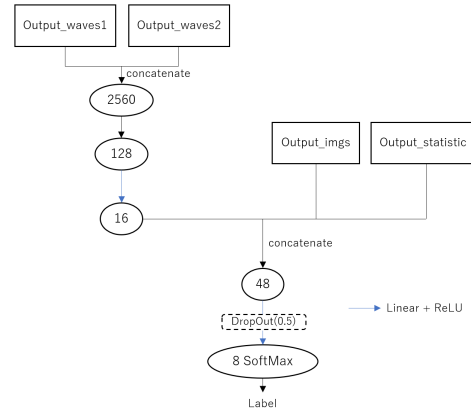


Figure 4: The model structure to predict modes of locomotion and transportation.

Input: 3.2.1

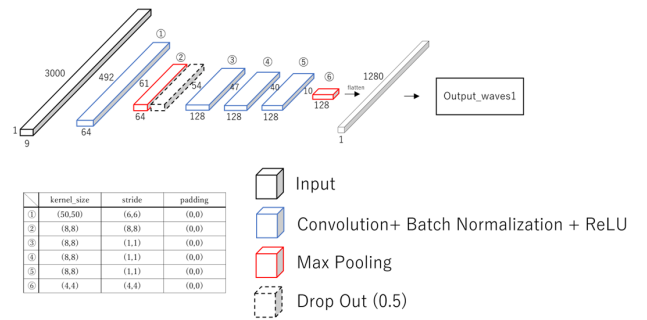


Figure 5: Output_waves1 section's model.

Input: 3.2.1

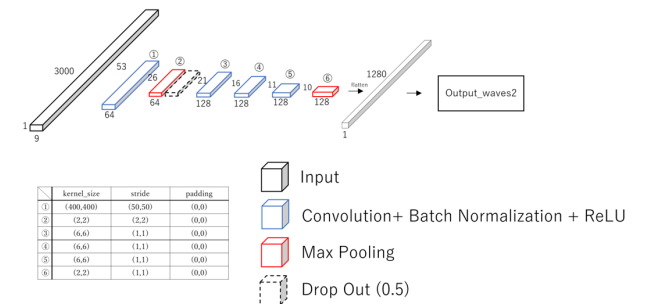


Figure 6: Output_waves2 section's model.

The accuracy for “run” state estimation is 99.8 % for train data and 95.7 % for validation data. In addition, although we expected that it would be difficult to differentiate between “train” and “subway”, these states were predicted unexpectedly well. However, “train,”

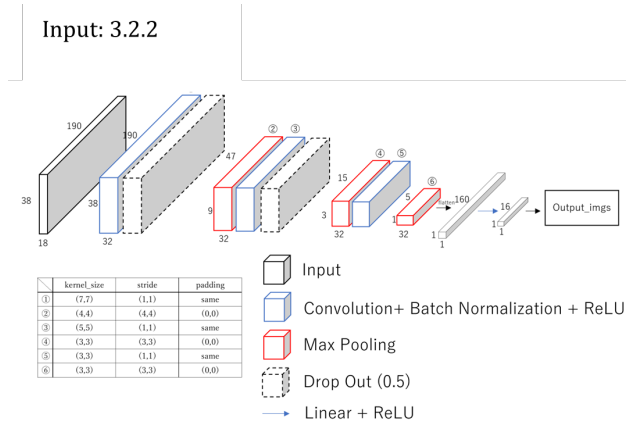


Figure 7: Output_imgs section's model.

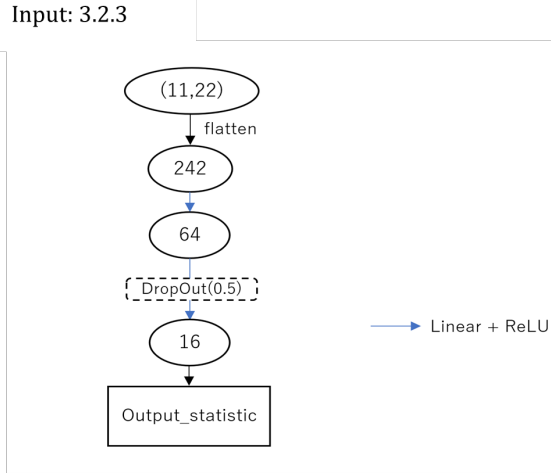


Figure 8: Output_statistic section's model.

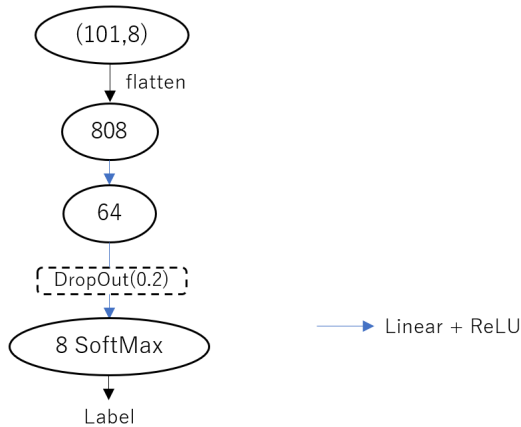


Figure 9: Postprocessing model structure to predict modes of locomotion and transportation for final evaluation.

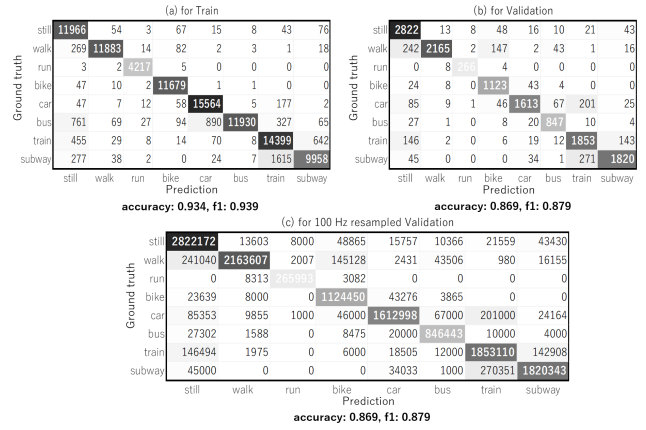


Figure 10: (a) Main processing result of train, (b) validation, and (c) 100 Hz resampled validation.

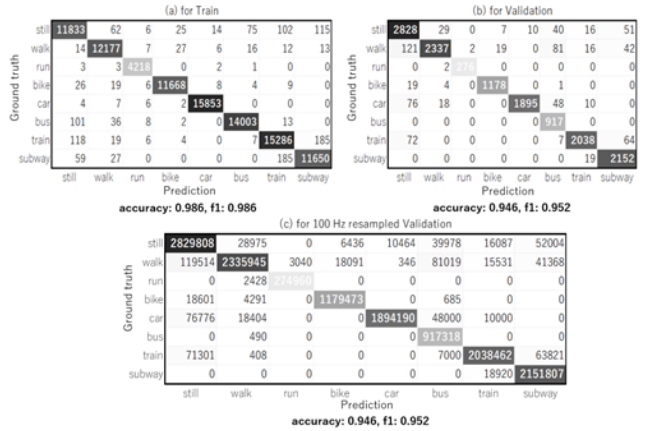


Figure 11: (a) Postprocessing result of train, (b) validation, and (c) 100-Hz resampled validation.

“bus,” and “subway” were frequently misclassified as the “still” state. Thus, it may be difficult to effectively differentiate between the stopped “train,” “bus,” and “subway” states and the “still” state. We found that the F1-score for the validation data every 10 s and the 100-Hz resampled validation were 99 % same: thus, no negative influence was observed when performing the estimation using the validation data every 10 s.

6.2 POSTPROCESSING RESULTS

The postprocessing results were also evaluated in terms of accuracy, F1-score, and confusion matrices. The results obtained on the training data, validation data, and 100-Hz resampled validation data are shown in Figure 11(a), Figure 11(b), and Figure 11(c), respectively.

6.3 EFFECT OF POSTPROCESSING

The transitions of the predicted labels prior to postprocessing, the predicted labels after postprocessing, and the correct labels for the validation data are shown in Figure 12.

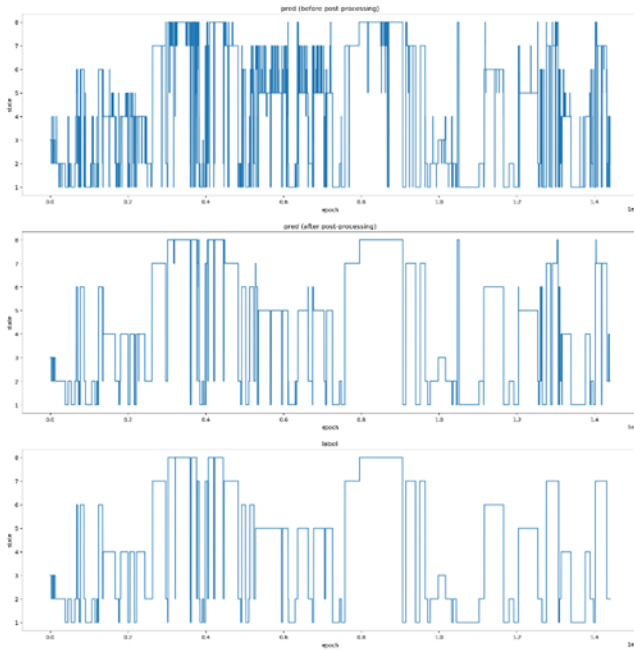


Figure 12: Transitions of three types of labels. (upper) Outputs of Main processing (middle) Outputs of Post processing (lower) Correct labels

By using the outputs of a large number of blocks, the state changes over extremely short periods were eliminated, and stable estimation results were obtained. Compared to before postprocessing, the incorrect estimation of the “train,” “bus” and “subway” states for the “still” state was reduced greatly. In particular, the probability of incorrect estimation in validation data reduced from 2.9 % to 0.0 % for “bus” state, from 6.7 % to 3.3 % for “train” state, from 2.1 % to 0.0 % for “subway” state. The “still” estimation for the stopped “bus,” “train” and “subway” was filled in by the state estimation around the stopping time, thereby resulting in correct estimation results.

7 CONCLUSION

In this research, long-term data were used to estimate modes of locomotion and transportation. The main model is a multi-input model with four input sections to estimate the motion state. In addition, postprocessing model was constructed to obtain the final

prediction in consideration of the previous and next states. As a result of these processes, the F1 score for the validation data was 0.952. Note that the recognition result for the testing dataset will be presented in the summary paper of the challenge [13].

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